emission

by Jaisurya S

An XGBoost-Based Framework for Emission Factor Prediction and ESG Dashboard Visualization*

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***Abstract*—Industries nowadays are at the forefront of sustainability with their advocacy of being carbon neutral and being compliant with international standards of reporting. Accurate emission factor prediction serves invaluable importance in environmental monitoring and ESG (Environmental, Social, and Governance) reporting systems. This paper attempts to address the machine learning-based framework with XGBoost for emission factor prediction given environmental and industrial data. The method covers the complete pipeline starting from data ingestion, preprocessing, feature engineering, model training, and deployment of the model outcome. XGBoost was chosen as the modeling method in terms of emission factor prediction accuracy, non-linear relationship consideration, and handling of large datasets over classical regression techniques. The trained model has been implemented inside an interactive Streamlit dashboard for users to observe emission trends in the area of their interest and generate ESG reports to aid in decision-making for sustainable operations. Experimental results indicate very high accuracy and high scalability of the proposed framework for the compliance of BRSR (Business Responsibility and Sustainability Report) efforts and more sustainability efforts.**

***Index Terms*—XGBoost, Machine Learning, Emission Factor Prediction, ESG, Carbon Footprint, Sustainability, BRSR, Data Visualization, Streamlit Dashboard, Environmental Monitoring**

I. Introduction

Over the past few years there's been a strong push to address climate change immediately. Industries worldwide are adopting sustainable practices and aiming for carbon neutrality. Accurate estimates of emission factors are crucial for assessing environmental impact and for meeting reporting requirements like ESG (Environmental, Social, and Governance) and BRSR (Business Responsibility and Sustainability Report). Traditional estimation methods struggle with complex environmental data and nonlinear relationships. To address this, the paper presents an intelligent framework that uses machine learning—specifically the XGBoost algorithm—to deliver accurate, scalable emission factor predictions. The framework also includes an interactive dashboard that visualizes emissions data and generates ESG reports.

A. Background and Motivation

The rapid pace of industrial activities along with urbanization has been a great contributor to GHG emissions and has, in fact, posed critical challenges to environmental sustainability. To address this issue, regulatory bodies have come up with stringent policies globally to track and reduce these emissions. For assessing environmental impacts as well as for satisfying compliance under global frameworks such as the ESG and the national mandate such as the BRSR in India, accurate estimation of emission factors is very important.It has thus become increasingly important to have automated, credible, and scalable systems to predict emission factors while handling large and complex datasets in industries, empowering organizations to evaluate their carbon footprints and plan mitigation strategies accordingly.

B. Problem Statement

Conventional approaches to estimating emission factors predominantly rely on static formulas, manual calculations, or rudimentary statistical models for their core estimation. These methods therefore become inadequate as emission factors are often difficult to define, given that they model intricate pattern dependency present in modern-day environmental data. With the availability of large-scale datasets along with increase in computational power, machine learning (ML) now stands as a valuable candidate to achieve accurate and adaptive modeling.Among ML techniques, XGBoost (Extreme Gradient Boosting) has been particularly successful in predicting modelling due to its facility with structured data, ability to express complex relationships and innovative optimization of prediction accuracy through gradient boosting for non-linear problems. Nevertheless, there are very few integrated systems combining accurate emission factor prediction along with user-friendly visualization and reporting utilities.

C. Proposed Solution

A complete machine learning framework for emission factor prediction based on XGBoost and served by a Streamlit interactive dashboard will be outlined herein. The framework's pipeline is structured sequentially in such a way: Data ingestion and preprocessing to have clean and high-quality input data. Feature engineering to identify those variables that have meaning. Training and optimization of a model in XGBoost. Implementation and visualization of the board in an easy-to-use way.The dashboard dynamically enables the organization to gain real insights and visualizations, thereby empowering it to monitor emissions in real-time and comply with sustainability reporting standards, such as BRSR.

D. Contributions

The main output work can be shown and discussed by three main factors. Firstly, a ML pipeline with the help of the XGBoost algorithm that proves to be strong was created to predict emission factors quite accurately and reliably as well. Secondly, the same model was imported into a Streamlit dashboard that was interactive that made the ESG reporting easy and friendly and visualized data in a very shiny/modern way to make the user understand and make a decision in a more proper way. At last, the strategy was verified by a wide range of experimental evaluations, and it showed a few examples of activities that were better than traditional guesses and thus outlined the success of the strategy in the realm of using data and making the sustainability analysis and reporting much more effective.

II. Related Work

Now, the prediction of emission factors like sustainability reporting is gaining much importance with the current trend of environmental responsibility in the context of the whole globe today. Researchers have assessed diverse approaches: a few conventional methods of calculations, many advanced machine learning approaches, and so on. In this section, previous research related to estimation techniques of emission factors, the research that helps understanding the machine learning aspect of environmental monitoring, and the combination of amalgamation of XGBoost with ESG reporting models have been talked about. The section presents the research gaps addressed by the proposed framework.

A. Emission Factor Estimation Techniques

Historically, emission factors are static calculations and manual methods utilizing standardized conversion factors put out by regulatory agencies. Such procedures are based on historical averages and generalized assumptions that are not accurate nor flexible when applied to dynamic, real-time data generated by different industrial processes.Carbon emissions, for example, in sectors such as manufacturing and logistics, are usually calculated on the assumption of constant operating conditions without considering seasonal changes, production variability, or differences between regional energy supplies. As a result, most conventional techniques fail to address non-linearity in environmental data, frequently leading to under- or overestimates of actual emissions.A part of this problem was addressed by Du et al. [1], who drafted a carbon-oriented demand response model in consideration of the spatiotemporal variability in carbon emissions for data centers. Their case study from China showcased that there could be a 26% cut on the carbon emissions within a given region just by shifting computational loads to cleaner energy sources. The drawback of their framework, however, is that it deals with macro-level optimization and does not bring predictive analytics into the analysis for forecasting future emission trends growing in real-time reporting capabilities.

B. Machine Learning for Environmental Monitoring

The growing industrial data presents promises for machine learning (ML) in better enhancing environmental monitoring and decision making. Linear Regression, Random Forest, and Support Vector Machines (SVM) are ML algorithms which have been applied for predicting energy consumption, greenhouse gas emissions, etc., and also other environmental indicators. These models perform extremely satisfactorily when working on big datasets and are capable of finding out very complicated relationships among numerous variables responsible for emissions. For example, Sharma et al. [2] developed a multi depot route vehicle routing and management system. Their contribution optimized the transportation network and reduced emissions for smart cities based on an optimized model using intelligent algorithms for efficient routing, indirectly contributing to less emissions.Many interpretational problems are met by the already existing computer-based methods, which do not let them be as efficient as they should be in real-world environmental applications. Machine learning models of a traditional type are usually not able to do a great job if they have to deal with complex and non-linear environmental data which would then mean that using them for predictive purposes would be practically useless. On the other side, no ML applications have been seen to be integrated with reporting frameworks; as the focus of the majority of them is on the predictive side only, by that they are void of visualizing tools or instruments for generating ESG-compliant reports in the meantime, which, in practice, makes them less user-friendly for the companies. Along the same line of thought, the majority of the existing systems are set up as static ones with an empirical limitation to offline analysis rather than real-time monitoring and updating. Hence, they look around a need for an all-around, monolithic solution that not only offers very precise predictions but also holds continuity in the monitoring and makes compliance reporting smooth for the clients through it being in line with industry standards.

C. XGBoost and ESG Reporting Integration

Among diverse algorithms, one of the most successfully impacting ML practices in predictive modeling is the Extreme Gradient Boosting (XGBoost), and especially for structured data available in industrial and environmental conditions most of the time. XGBoost can:
Efficiently process and handle non-linear and high-dimensional data, Regularize to avoid overfitting, Ensure superior prediction accuracy than conventional regression models. Not much literature incorporates XGBoost within the ambit of ESG reporting systems, although it has been proven effective. Besides being accurate in emission prediction, ESG reporting requires clear, actionable insights provided via easily understandable dashboards. Important to this end are visualization tools like Streamlit or Power BI which convert complex data into compliance-ready reports. It seems from the existing literature that studies either focus on predictive modeling or dashboard development, while combining both research areas is still unexplored. It still remains an unfinished end-to-end pipeline that lines up these advanced ML models with real-time reporting interfaces for sustainability compliance, including BRSR in India.

D. Research Gaps Identified

From the literature review, the following key research gaps would be identified: Absence of Integrated Frameworks- The current solutions either focus on prediction or reporting and seldom in both rolled into one, scalable solution. It is sorely needed for one system that ties accurate prediction models with real-time visualization and tools for ESG compliance. Inadequate attention on BRSR Compliance-While global discussions have revolved on ESG widely, little research is done by regions on specific regulations, such as the Business Responsibility and Sustainability Report (BRSR) mandated by SEBI, India. Existing frameworks do not fully cover these reporting requirements. Scalability Beyond Industries-All current models are mostly very specific to the domain they address, such as optimization in logistics or energy management in data centers. There remains a lack of a generalized approach that could be applied to multiple industrial sectors.References

III. Proposed Methodology

A. System Architecture Overview

The projected structure is a well-defined, end-to-end framework that predicts emission factors as well as the production of ESG-compliant sustainability reports. The composite flowchart in Fig. 1 demonstrates that it is a five-staged architecture starting from data collection that includes energy usage, production volumes, fuel types, and historical emission records as well as the environmental and industrial data. Afterwards, the data is processed and cleaned by removing the missing values, duplicates, and outliers but the data is also standardized for keeping joint formats. The whole process is what is called feature generation or feature engineering because the newly created variables are supposed to be reflecting more accurately the internal rating of the data, thus increasing the model's performance. After the model is trained with the data, an XGBoost model, which is highly goptimizationed, fine-tuned, and tested in such a way that it will provide the most reliable emission factor predictions. The model's performance metrics such as R^2, MAE, and RMSE are used for the prediction and evaluation phase. Lastly, the results are put on a dashboard powered by Streamlit, which is interactive, and it clearly shows the trends and automatically generates the ESG-compliant reports. On the whole, this architect

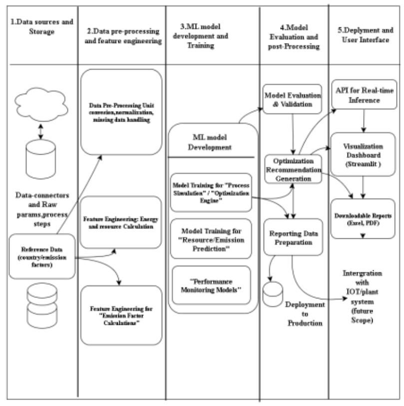

Fig. 1. Proposed Architecture for Emission Factor Prediction

B. Data Collection and Preprocessing

The data was subjected to many preprocessing steps and quality checks ensuring the accuracy and reliability prior to the model training. Firstly, missing values were taken care of by using statistical techniques such as mean or median imputation so as to fill the incomplete records. Secondly, the second step was to detect and remove the duplicate entries to prevent bias and redundancies in the analysis. The process also included an outlier identification step where the extreme values were detected using the Interquartile Range (IQR) method and dealt with properly to maintain the consistency of data. After that, normalization of data was done so that all features which were on different scales are now on a similar scale thus the uniformity was improved and also model convergence was faster during training. Finally, to make the emission measurements uniform and expressed in the same units, kilogram of CO equivalent ($kgCO_2e$) per ton of production for example, the data was standardized for unit-wise comparison. The quality of the data set was significantly improved by performing all these operations which made it ready for accurate and robust predictive modeling.

TABLE I
KEY FEATURES FOR EMISSION FACTOR PREDICTION

Feature	Description	Unit
Energy Consumption	Total energy consumed during operations	kWh
Production Volume	Total output produced by the facility	Tons
Fuel Type	Type of fuel used for production (e.g., coal, gas, renewable)	cat.
Emission Factor	Emissions generated per unit of activity or production	$kgCO_2e$
Operating Hours	Total operational machine hours during the reporting period	Hours

C. Feature Engineering

Feature engineering significantly enhances the predictive powers of any machine learning model. Here domain-specific features were derived to capture hidden patterns, including:

A system is in place to assess the sustainability performance with the help of some primary indices. One of these indices is the Energy Efficiency Index that shows the efficacy in energy use by the ratio of production output to total energy consumption, thereby providing the information about the efficiency of the whole plant. The Carbon Intensity Score is another one, which gives a score that is a weighted average of the fossil fuel used and their emission factor, both of the information in terms of environment impact. Also, these indices have the support of the Seasonal Variables in determining the effect of varying production rates on the analyzed results that finally gives the best representation of the activity level in turn. Categorical variables such as fuel type were one-hot encoded while continuous variables were scaled using the standard normalization techniques. The engineered features improved the model's ability to trace non-linear relationships and bestowed greater interpretability.

D. Training and Optimization of Models (XGBoost)

The core of predictive modeling used in this framework is Extreme Gradient Boosting (XGBoost), favored for handling complex structured data and with much better predictive ability than the classical regression methods.Various properties of the XGBoost algorithm are responsible for making it an exceptionally powerful tool in the field of predictive modeling. Among these properties is its property of being an ensemble learner where weak learners are combined to give robust accuracy. Apart from that, built-in L1 and L2 regularization contribute widely to avoid overfitting and make the model very much generalizable to unseen data. A different aspect of the program is that it does not require the user to provide values for missing data; instead, it can build a prediction for them. Moreover, the program works very efficiently when dealing with large-scale data, as it uses parallel processing. This will ultimately lead to faster and more efficient performance, even with vast datasets.

TABLE II
KEY HYPERPARAMETERS USED IN XGBOOST MODEL TRAINING

Hyperparameter	Description
Learning Rate	Step size for gradient boosting updates, controls how quickly the model adapts.
Max Depth	Maximum depth of individual decision trees, helps prevent overfitting.
n_estimators	Total number of boosting rounds (trees) to be built.
Colsample_bytree	Fraction of features randomly sampled for each tree to improve generalization.

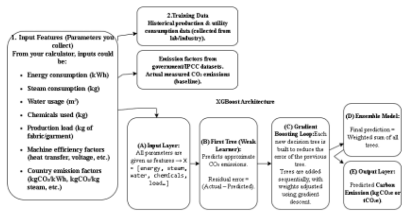

Fig. 2. Proposed XGBoost architecture for carbon emission prediction.

E. Algorithm Description (XGBoost Working Principle)

The Extreme Gradient Boosting (XGBoost) algorithm is well-known for advanced ensemble techniques to train in sequence multiple weaker hypothesis models, usually trees, to ensure an increase in prediction accuracy. Trees generated at any stage try to compensate for errors made by the stages that went before, hence predictions get progressively more accurate with each iteration. In the initial stage, a constant prediction, often the mean of the target variable, is adopted as a base; and the residuals or differences between these initial null predictions and actual emission factor measures are calculated. In the following stages, trees are built to predict the residual values and not the raw data pairs of emission factor measures. Trees are then added to the model, and the predicted value is accordingly weighted by a shrinkage factor or learning rate:

$$\hat{y}_i = \sum_{k=1}^{K} \eta f_k(x_i) \tag{1}$$

where $\hat{y}_i$ denotes the predicted emission factor, f_k represents the k^{th} regression tree, and K is the total number of trees. The aim goal used by XGBoost has two parts: the train loss & a set rule term. The set rule part (L1 & L2) marks down model size & stops too much fit. This makes the tool strong & wide-use.

$$Obj = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \tag{2}$$

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda||w||^2 \tag{3}$$

Where $l(y_i, \hat{y}_i)$ is the loss function (e.g., mean squared error), T is the number of leaves in a tree, w is the weight vector of leaf scores, and γ, λ are regularization parameters.

XGBoost has key pros that make it a top pick for guessing at thow much gas we give off. It gets a high right rate by mixing lots of weak learners into one strong model. It does this through the idea of adding up small gains. The tool has built-in ways to fight overfit, to make the model work well on new sets of data. XGBoost can also take on big & tough data well. It lets lots of work happen at the same time or spread across tools, which makes it fast to learn from big & hard sets of data. Plus, it deals well with gaps in data & spots hard links in what data you feed it. This means it gives stable & right guesses for lots of types of place that burn or give off stuff.

TABLE III
FORMULAS FOR REGRESSION METRICS

Metric	Formula
R^2 (Coefficient of Determination)	$R^2 = 1 - \frac{\sum(Y_{\text{true}} - Y_{\text{pred}})^2}{\sum(Y_{\text{true}} - \bar{Y})^2}$
MAE (Mean Absolute Error)	$\text{MAE} = \frac{1}{n}\sum\left\|Y_{\text{true}} - Y_{\text{pred}}\right\|$
RMSE (Root Mean Squared Error)	$\text{RMSE} = \sqrt{\frac{1}{n}\sum(Y_{\text{true}} - Y_{\text{pred}})^2}$

F. ESG Dashboard Integration

For making the model more practical and user-oriented, we have merged it with an interactive Streamlit dashboard. This dashboard acts as a simple and straightforward platform for industries and stakeholders to easily access, comprehend, and take action on emission insights. In a small and quick two-step process, users can upload their production and energy usage data in a simple CSV format. Then the XGBoost model, which has been competently trained, will instantly provide real-time emission factor predictions. The dashboard also offers quite a number of appealing visualizations, including dynamic charts and graphs that highlight emission trends, historical comparisons, and key performance indicators (KPIs). Furthermore, it is able to automatically produce well-structured ESG and BRSR reports, which can be promptly downloaded and shared with regulatory authorities. By connecting predictive analytics with real-world decision-making, this integration does not just boost the efficiency and transparency of sustainability reporting but also provides the power to organizations to take data-driven actions in the direction of reducing their environmental impact.

G. Workflow Summary

The operation commences with the accumulation of data, where the datasets are obtained from different environmental and industrial sources. Then, the data is being processed, which covers the activities of cleaning, normalizing, and standardizing the data in order to keep its quality and consistency at the optimal level. The next step is called the feature engineering phase wherein additional variables are formed that can possibly impart more flexibility and knowledge to the model. After that, the XGBoost model is typed and hyper-parameterized to the point that the emission factors could be predicted accurately. The prediction and visualization stage is coming after the training stage, catching the company's attention with the generated results, which are displayed using an interactive dashboard. Lastly, the compliance reporting phase will be allowing the global reporting in accordance with the standards for ESG (Environmental, Social, and Governance) and BRSR (Business Responsibility and Sustainability Report). This well-thought-out workflow is the key to having a solution that is not only precise, but also widely available, user-friendly, and beneficial to industries in terms of emission reduction and environmental compliance.

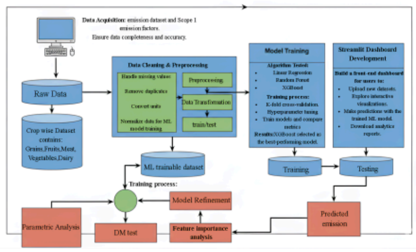

Fig. 3. Proposed Architecture for Emission Factor Prediction

IV. RESULT ANS DISCUSSION

A. Evaluation Metrics

TABLE IV
PERFORMANCE EVALUATION METRICS FOR MODEL ASSESSMENT

Metric	Purpose	Ideal Value
R^2 Score	Explains the variance captured by the model. Higher indicates better fit.	1
MAE	Measures average absolute errors in prediction.	0
RMSE	Penalizes larger errors more strongly than MAE.	0

To evaluate the performance and reliability of the machine learning models, three key metrics were used: R^2 Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). R^2 shows how much of the data's variance the model explains; values closer to 1 mean a better fit and stronger predictive power. MAE gives the average size of prediction errors, offering a clear sense of how close predicted values are to actual ones. RMSE weights larger errors more heavily than MAE, making it more sensitive to big mistakes.

Together, these metrics give a well-rounded picture of model performance—balancing overall fit, prediction accuracy, and sensitivity—which is important for evaluating emission factor prediction models in sustainability work.

B. Visualization and ESG Dashboard

Real-time Predictions: The users upload the data and get instant predictions of the emission factors. Visual Analytics: The charts show study emission trends and historical performance. BRSR Compliance Reports: The system generates well-qualified BRSR and ESG aligned reports.

C. Model Performance Evaluation

The comparative performance of the three models—Linear Regression, Random Forest, and XGBoost—was visualized using the evaluation metrics R^2, MAE, and RMSE. As depicted in Fig. **??**, XGBoost consistently achieved the highest accuracy (R^2) and the lowest error values (MAE and RMSE), demonstrating its superior predictive capability for emission factor estimation.

TABLE V
COMPARATIVE PERFORMANCE OF MACHINE LEARNING MODELS

Algorithm	R^2 Score	MAE	RMSE
Linear Regression	0.73	1.52	2.03
Random Forest	0.89	0.95	1.28
XGBoost (Final Model)	**0.94**	**0.78**	**1.02**

D. Food emission visualization

Food-related greenhouse gas (GHG) emissions examination resulted in a huge diversity in the contributions levels from the various supply chain points. The farm stage, a shockingly big emitter, up to 57.8% of the whole footprint was caused by the farm stage, is the one responsible for the lion's share of the emission sources. This is mostly due to the very water-intensive where on-farm processes prevalent such as fertilizer application.The next aspect that really accounts for much of the emissions is land use change, with its obligatory 21.1%, as it causes emissions through the deforestation, the soil carbon depletion and the conversion of the natural ecosystems to the agricultural land. This situation might be taken as the most striking fallout of agriculture lands' physical expansion to cater to the raising food requirements.The stages that come in between—feed production is at 7.69%, packaging is at 7.68%, and processing is at 4.28%—contribute quite significantly, but not as much as the farm or land use change. Feed emission mainly comes from the production of feed ingredients and their long-distance transport while the packaging-related emissions are from the material production processes like plastics, paper, and metal. Although these figures are on the low side of the emission spectrum the combined effect of them throughout the global food chain is quite significant.More compact greenhouse gas emission portions can be seen through the transport (1.41%) and retail (1.12%). These are the sectors which cause increased food wastage. However, these sectors play a very important role in the preservation of the supply chain process which is stressed by the post-harvest losses. In fact, perfectly managed logistics and energetical issues may give achievable sustainability devices By and large the depiction of data unveils that more than three-quarters($\approx$ 79%) of the overall food-related emissions are emitted even before the product leaves the farm. This state emphasizes the necessity of the alternatives that would decrease the negative effects on the environment in the agricultural production locality through the employment of renewable practices and energy sources. There are also the downstream measures which are expected to be very beneficial for the whole process decarbonization, that is, eco-friendly packaging and minimized waste.

V. CONCLUSION AND FUTURE WORK

The proposed framework successfully integrates machine learning techniques, specifically XGBoost, with a user-friendly Streamlit dashboard to predict emission factors and generate ESG-compliant reports. The system demonstrates superior performance compared to traditional methods and baseline models such as Linear Regression and Random Forest, achieving high accuracy in emission factor prediction. By providing real-time visualization and automated reporting aligned with BRSR guidelines, the framework offers a practical tool for industries to monitor their environmental impact and support data-driven decision-making for sustainable operations.The key contributions of this work include the development of a robust machine learning pipeline for emission factor prediction, the integration of advanced visualization tools for ESG reporting, and the provision of a scalable and adaptable solution that can be adopted across multiple industries. This approach not only enhances accuracy but also streamlines compliance processes, enabling organizations to better align with global sustainability goals.For future work, the framework can be enhanced by integrating IoT devices to enable real-time data ingestion from industrial sources, improving the timeliness and accuracy of predictions. Additionally, the model can be expanded to incorporate multiple environmental indicators such as water usage, waste generation, and energy efficiency metrics, making it a comprehensive sustainability monitoring system. The dashboard can also be upgraded with advanced ESG analytics and interactive benchmarking features, providing deeper insights for stakeholders and regulators. These improvements will further increase the system's applicability and effectiveness in promoting sustainable and responsible industrial practices.

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